But now he would be a bird on his own, no longer part of a relatively anonymous flock, but an odd man out. Hawks are actually known to go for odd pigeons out, but even if this were not so there are plenty of theoretical reasons for thinking that breaking ranks might be a suicidal policy. Even if his companions eventually follow him, the individual who first flies up off the ground temporarily increases his domain of danger. Whether Hamilton's particular theory is right or wrong, there must be some important advantage in living in flocks; otherwise, the birds would not do it. Whatever that advantage may be, the individual who leaves the flock ahead of the others will, at least in part, forfeit that advantage. If he must not break ranks, then, what is the observant bird to do? Perhaps he should just carry on as if nothing had happened and rely on the protection afforded by his membership of the flock. But this too carries grave risks. He is still out in the open, highly vulnerable. He would be much safer up in a tree. The best policy is indeed to fly up into a tree, *but to make sure everybody else does too*. That way, he will not become an odd man out and he will not forfeit the advantages of being part of a crowd, but he will gain the advantage of flying off into cover. Once again, uttering a warning call is seen to have a purely selfish advantage. E. L. Charnov and J. R. Krebs have proposed a similar theory in which they go so far as to use the word 'manipulation' to describe what the calling bird does to the rest of his flock. We have come a long way from pure, disinterested altruism!

Superficially, these theories may seem incompatible with the statement that the individual who gives the alarm call endangers himself. Really there is no incompatibility. He would endanger himself even more by not calling. Some individuals have died because they gave alarm calls, especially the ones whose calls were easy to locate. Other individuals have died because they did not give alarm calls. The *cave* theory and the 'never break ranks' theory are just two out of many ways of explaining why.

What of the stotting Thomson's gazelle, which I mentioned in Chapter 1, and whose apparently suicidal altruism moved Ardrey to state categorically that it could be explained only by group selection? Here the selfish gene theory has a more exacting challenge. Alarm calls in birds do work, but they are clearly designed to be as inconspicuous and discreet as possible. Not so the stotting high jumps. They are ostentatious to the point of downright provocation. The gazelles look as if they are